Introduction

The word 'computer' usually conjures up images of a desktop, a laptop, a server or perhaps if you're a really big thinker, even a mainframe. In reality computers are not just dumb boxes which just sit around on tables. As a matter of fact, computers are embedded into everything ranging from your microwave to the Boeing 777, your smartphone to the Curiosity Mars Rover. All in all we're surrounded on all sides by these small wonders; about 90% of the processors manufactured across the world are powering what are also called embedded systems. Such systems despite being small and seemingly non-existent are the reason for the rapid growth and advancement in other areas of technology.

These devices are responsible for powering everything from mundane home appliances, traffic lights and simple robots to bewilderingly complex systems deployed for process and production control at automated manufacturing plants, power stations and telecommunication systems. They're also used heavily in avionics and air traffic control. Apart from this, embedded systems find use in space applications – Space X Dragon, Mars Pathfinder and Curiosity all carried such systems.

This FastTrack will delve into the workings of such Embedded Systems. We'll start with an introduction to embedded systems and why are they important followed by their functions and the scope of their applications. Then we'll take up, hardware – design principles, layout considerations and component selection along with the challenges that such systems throw at designers due to constraints on power consumption, heat dissipation and real time performance. On the software side we'll talk about how software is written for such systems and how it differs from non-embedded software. We'll also discuss the operating systems employed on such devices and their functioning. Finally we'll talk about the future with mobile and ubiquitous computing, how will we interact with computers which are embedded right into our environment and how will the revolutionize the way in which we interact with it.

Of course there's no way this little booklet will be able to cover every aspect of such systems in detail but we've tried to give you a birds eye view of how such systems function and hopefully spark your curiosity and spur you onwards into delving deeper into this micro-universe. We hope you enjoy reading this FastTrack as much as we enjoyed writing it. As always, happy reading.